Electric Flux and Gauss's Law

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Electric Flux and Gauss's Law

Electric Flux

Electric flux through a surface is defined as the dot product of the electric field vector and the area vector of that surface.

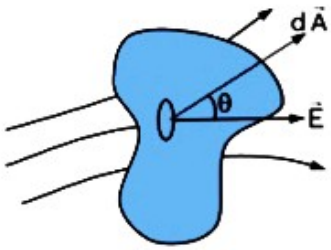
It is a measure of the number of electric field lines passing through that surface.

Electric Flux passing through an open surface

Electric flux passing through an elemental area dA,

$d\phi = EdA\cos\theta$

$d\phi = \vec{E}.d\vec{A}$



Here $\vec{A}$ is directed perpendicularly to the either side of the open surface depending on our interest.

Here $dA.\cos\theta$ = Effective or projected area along $\vec{E}$,

$E\cos\theta$ = Effective field or field component in the direction of $d\vec{A}$.

θ = Angle between $\vec{E}$ and $d\vec{A}$

Electric flux passing through the whole surface,

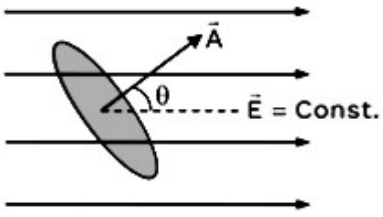
$$\phi = \int \vec{E}.d\vec{A}$$

Special Cases

When electric field $\vec{E}$ and θ is constant,

By definition, electric flux passing through shaded area is,

$$\phi = \vec{E}.\vec{A} = EA\cos\theta$$



Electric flux is a physical quantity and is proportional to the number of field lines (imaginary lines) crossing through the area of our interest.

ϕ is a scalar quantity with unit $N\text{-}m^2/C$ or V-m (gauss). ϕ can be +ve, –ve or zero.

For $0° \le \theta < 90°$,

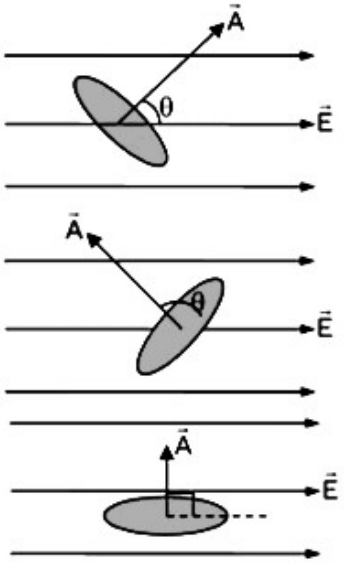
$\phi > 0$.

For $90° < \theta \le 180°$,

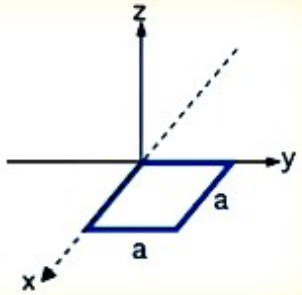
$\phi < 0$.

For $\theta = 90°$,

$\phi = 0$.



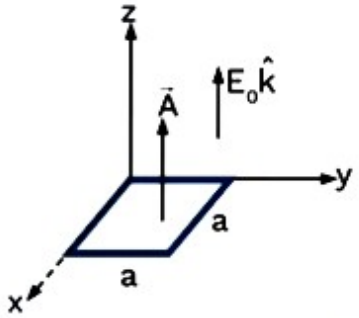
Q. **Find flux of following electric field $E_0\hat{k}$ through the area shown.**



Sol: Electric flux through the area, $\phi_{area} = \vec{E}.\vec{A}$

$$\phi = (E\hat{k}).(a^2\hat{k})$$

$$\phi = E_0 a^2$$



Q. **If an electric field is given by $\left(10\hat{i} + 3\hat{j} + 4\hat{k}\right)$, then calculate the electric flux through a surface of area 10 units lying in y–z plane.**

Sol: As the surface lies in y-z plane, so the area vector will be perpendicular to it along x direction.

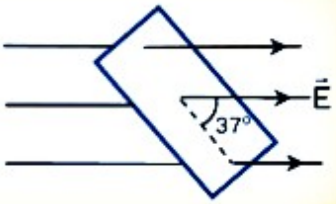
$$\phi_{surface} = \vec{E}.\vec{A}$$

$$\phi = \left(10\hat{i} + 3\hat{j} + 4\hat{k}\right).\left(10\hat{i}\right)$$

$$\phi = 100 + 0 + 0$$

$$\phi = 100 \text{ units}$$

Q. **A flat rectangular plate with dimensions 4cm × 6cm is set with its plane at 37° to a uniform electric field E = 600 N/C as shown below. What is the flux through the plate?**



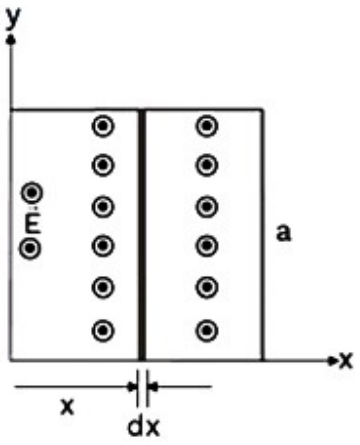
Sol: Area vector of a surface is inclined perpendicular to the plane of the surface. As the plane of the surface is making 37° with the electric field, so the area vector will make an angle of 53° with the electric field.

$$\phi = \vec{E}.\vec{A} = EA\cos\theta = 600 \times \left(4 \times 10^{-2} \times 6 \times 10^{-2}\right)\cos\left(90° - 37°\right)$$

$$\phi = 0.864 \text{ Nm}^2 / \text{C}$$

Q. **A square of side 'a' is lying in xy plane such that two of its side are lying on the axis. If electric field $\vec{E} = E_0 x \hat{k}$ is applied on the square, then the flux passing through the square is**

Sol: Here the electric field is variable. So, we have to apply the integration approach. Consider an elemental area as shown in the figure.



Flux passing through elemental area,

$$d\phi = \vec{E}.d\vec{A}$$

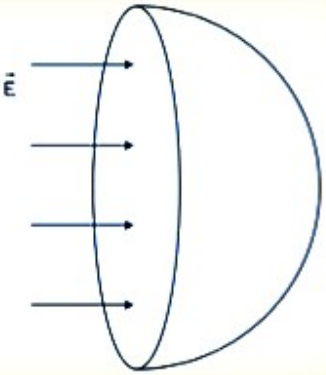
$$d\phi = (E_0 x).(a dx)\cos 0°$$

$$d\phi = E_0 (ax) dx$$

Total flux through the square,

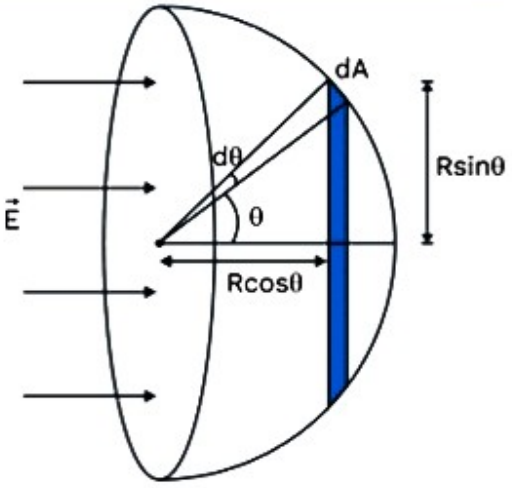
$$\phi = \int d\phi = E_0 a \int_0^a x\,dx \Rightarrow \phi = \frac{E_0 a^3}{2}$$

Q. **Find the electric flux passing through the curved surface of the hemisphere of radius R.**



Sol: **Method-1**

Consider an elemental area on the hemisphere as shown.



$$dA = (2\pi R \sin\theta).R d\theta$$

$$dA = 2\pi R^2 \sin\theta\, d\theta$$

Electric flux passing through dA,

$$d\phi = \vec{E}.d\vec{A} = E(2\pi R^2 \sin\theta\, d\theta)\cos\theta$$

$$d\phi = \pi E R^2 \sin 2\theta\, d\theta$$

Total flux passing through the curved surface,

$$\phi = \pi E R^2 \int_0^{\pi/2} \sin 2\theta d\theta$$

$$\phi = \pi ER^2\left[-\frac{\cos 2\theta}{2}\right]_0^{\pi/2} = \frac{\pi ER^2}{2}\left[(-\cos\pi)-(-\cos 0)\right]$$

$\phi = E\pi R^2$.

Method – 2

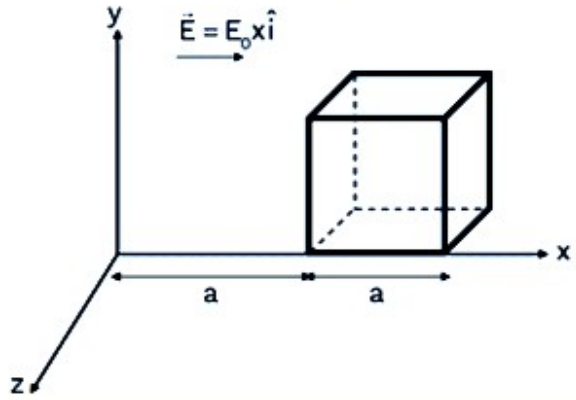
Effective area of hemisphere as seen from the electric field $\vec{E}$ is,

$$\vec{A} = \int dA\cos\theta = \pi R^2\,\hat{E}$$

Total flux,

$$\phi = \vec{E}.\vec{A} = E\left(\pi R^2\right)$$

Q. **Find the net flux passing through the cube in the figure given below.**



Sol: Electric flux through the left surface of the cube,

$$(\phi_{left})_{x=a} = \left(E_0 a\hat{i}\right).\left(-a^2\hat{i}\right)$$

$$\phi_{left} = -E_0 a^3$$

Similarly, $(\phi_{right})_{x=2a} = (E_0 2a)\hat{i}.\left(a^2\hat{i}\right)$

$$\phi_{right} = 2E_0 a^3;$$

$$\phi_{front} = \int \vec{E}.d\vec{A} = \int E\,dA\cos 90° = 0$$

$$\phi_{back} = \int E\,dA\cos 90^c = 0$$

$$\phi_{top} = \int E\,dA\cos 90° = 0$$

$$\phi_{bottom} = \int E\,dA\cos 90° = 0$$

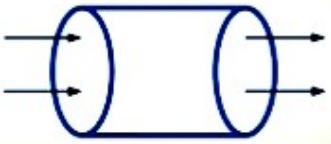
$\because \phi_{cube} = \sum \phi = \phi_{left} + \phi_{right} + \phi_{front} + \phi_{back} + \phi_{top} + \phi_{bottom}$

$\phi_{cube} = E_0 a^3$

Note:

If $\vec{E}$ = constant, then whatever be the orientation of closed surface (like cube, sphere, tube, etc), the net flux passing through it will always be zero.

Q. **A cylinder of radius R and length L is placed in a uniform electric field E parallel to the axis of the cylinder. The total flux through the cylinder.**



Sol:

$\phi_{cylinder} = \phi_{left} + \phi_{curvec} + \phi_{right}$

$\phi = E(\pi R^2)\cos 180° + \int EdA \cos 90° + E\pi R^2 \cos 0°$

$= -E\pi R^2 + 0 + E\pi R^2$

$\Rightarrow \phi = 0.$

Q. **Determine the net flux ϕ_N, entering flux ϕ_E and leaving flux ϕ_L through the hemispherical surface as shown.**

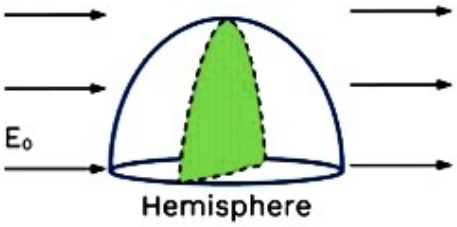


Sol: Effective area from the field direction

$A_{eff} = \frac{\pi R^2}{2}$

$\phi_{entering} = -\left(\frac{E_0 \pi R^2}{2}\right)$

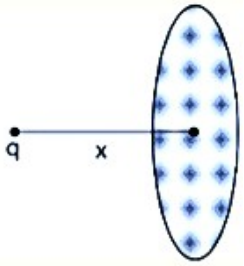
$\phi_{leaving} = \frac{E_0 \pi R^2}{2}$



Total flux $\phi_N = \phi_{entering} + \phi_{leaving}$

$\Rightarrow \phi_N = 0$

Q. **Find the flux passing through the given hypothetical disc. Charge q is present on the axis of the disc at a distance 'x' away.**



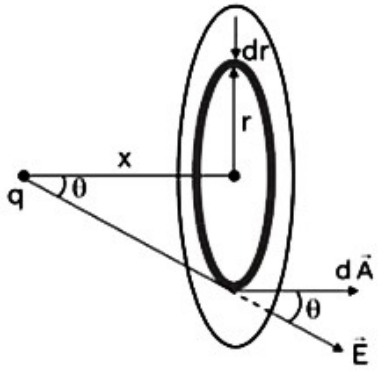
Sol: Flux passing through the elemental area dA on the disc,

$$d\phi = \vec{E}.d\vec{A}$$

$$d\phi = \frac{q}{4\pi\varepsilon_0\left(x^2+r^2\right)}\left(2\pi r dr\right)\cos\theta$$

Also, $\cos\theta = \frac{x}{\sqrt{x^2+r^2}}$

So, $d\phi = \frac{qx}{2\varepsilon_0}\frac{r\,dr}{\left(r^2+x^2\right)^{3/2}}$

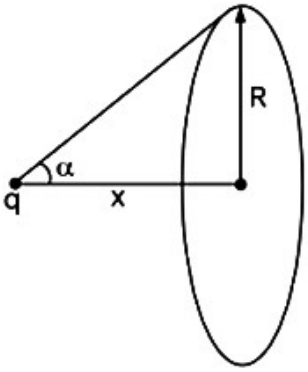


Total flux,

$$\phi_{disc} = \frac{qx}{2\varepsilon_0}\int_0^R \frac{r dr}{\left(r^2+x^2\right)^{3/2}} = \frac{qx}{2\varepsilon_0}\left[\frac{-1}{\sqrt{r^2+x^2}}\right]_0^R$$

$$\phi_{disc} = \frac{qx}{2\varepsilon_0}\left[\frac{-1}{\sqrt{R^2+x^2}}+\frac{1}{x}\right]$$

$$\Rightarrow \phi_{disc} = \frac{q}{2\varepsilon_0}\left[1-\frac{x}{\sqrt{R^2+x^2}}\right]$$

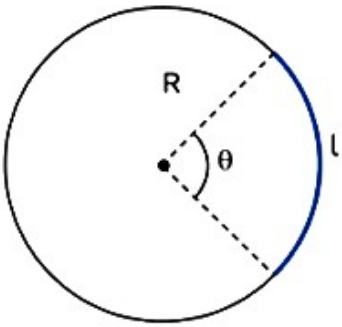


The result can also be written in following way.

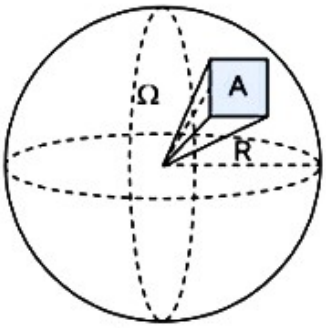
$$\phi = \frac{q}{2\varepsilon_0}\left(1-\cos\alpha\right)$$

where α is angle between x and R as shown in figure.

Plane Angle vs Solid Angle



$\theta = \frac{\ell}{R}$ Radians (rad)



$\Omega = \frac{A}{R^2}$ Steradians (sr)

Both plane angle (θ) and solid angle (Ω) are dimensionless.

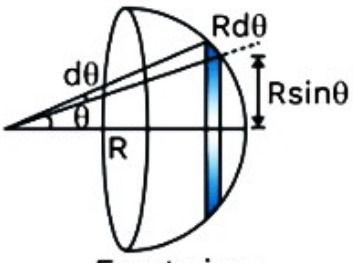
For a small arc on circle, $d\theta = \frac{d\ell}{R}$

For a small arc on sphere, $d\Omega = \frac{dA}{R^2}$

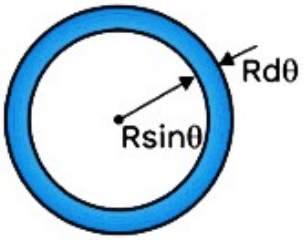
More specifically, $d\Omega = \frac{d\vec{A}.\hat{r}}{R^2}$

$d\Omega = \frac{dA\cos\theta}{R^2}$ where dA cosθ is effective area in the radial direction.

Relation between Plane Angle and Solid Angle



Front view



Side view

Solid angle made by the elementary ring (radius R sinθ, width R dθ) at the centre is

$$d\Omega = \frac{dA}{R^2} = \frac{2\pi R\sin\theta.Rd\theta}{R^2} = 2\pi\sin\theta.d\theta$$

Total solid angle corresponding to plane angle θ is

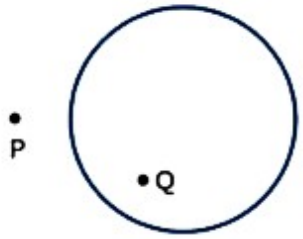
$$\int d\Omega = 2\pi\int_0^{\theta}\sin\theta d\theta$$

$$\Rightarrow \quad \Omega = 2\pi\left[-\cos\theta\right]_0^{\theta} = 2\pi\left(1-\cos\theta\right)$$

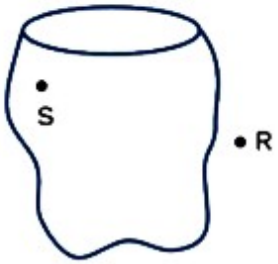
For $\theta = 90^\circ = \frac{\pi}{2}$ rad. $\Rightarrow \Omega = 2\pi$ sr. (hemisphere)

For $\theta_{max} = 180^\circ = \pi$ rad. $\Rightarrow \Omega_{max} = 4\pi$ sr. (sphere)

The total solid angle subtended by a closed surface at any point inside the surface is always 4π, regardless the size and shape of the closed surface.
On the other side, total solid angle subtended at any external point is always zero.



Sphere



Any closed surface

In the above figures, $\int d\Omega = 0$ and $\int_Q d\Omega = 4\pi$.

$\int_R d\Omega = 0$ and $\int_S d\Omega = 4\pi$.

Total electric flux produced by a charge

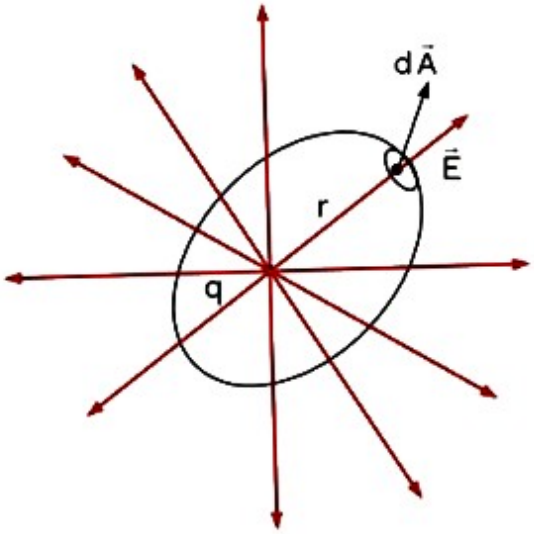
Any point charge q can be assumed to be enclosed in a closed surface as shown.

Flux passing through $d\vec{A}$,

$$d\phi = \vec{E}.d\vec{A} = (E\hat{r}).\left(dA\hat{r}\right) = E\,dA\cos\theta$$

$$d\phi = \frac{q}{4\pi\varepsilon_0 r^2} dA\cos\theta$$

$$d\phi = \frac{q}{4\pi\varepsilon_0}\left(\frac{dA\cos\theta}{r^2}\right) = \frac{q}{4\pi\varepsilon_0} d\Omega$$

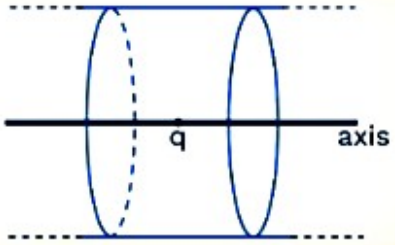


Total flux passing through the closed surface,

$$\int_S d\phi = \frac{q}{4\pi\varepsilon_0}\int_S d\Omega = \frac{q}{4\pi\varepsilon_0}.4\pi$$

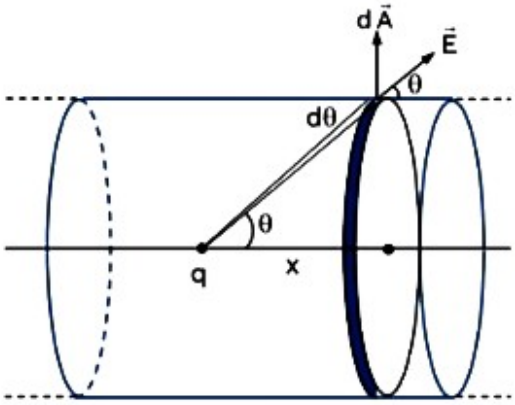
$$\phi = \frac{q}{\varepsilon_0}$$

Q. **A charge q is placed inside the infinite length cylinder of radius R. Find flux passing through curved surface.**



Sol: **Method-1**

Electric flux passing through the elemental ring of radius R and width dx is



$$d\phi = \vec{E}.\left(d\vec{A}\right) = E\left(dA\right)\sin\theta \ \cdots$$

(here the angle is 90-θ).

$$d\phi = \frac{q}{4\pi\varepsilon_0\left(x^2+R^2\right)}\left(2\pi R dx\right)\frac{R}{\sqrt{x^2+R^2}}$$

$$= \frac{qR^2}{2\varepsilon_0}\frac{dx}{\left(x^2+R^2\right)^{3/2}}$$

Total flux, $\phi = \frac{qR^2}{2\varepsilon_0}\int_{-\infty}^{+\infty}\frac{dx}{\left(x^2+R^2\right)^{3/2}}$

By solving above integration, we will get, $\phi = \frac{q}{\varepsilon_0}$.

Method-2

An infinite length cylinder always subtends 4π sr. solid angle at any internal point.

$\therefore\ \phi = \frac{q}{\varepsilon_0}$.

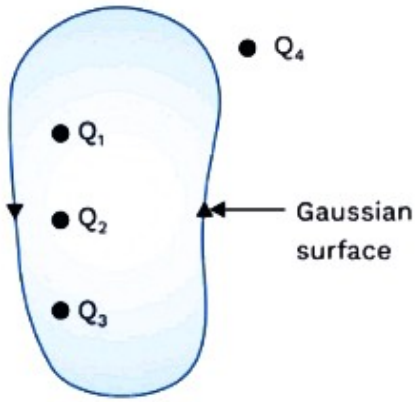
Q. **Find the electric flux coming out from the entire cube of edge a, at the centre of which a point charge q is placed.**

Sol: As the charge is enclosed within the cube, so total solid angle is 4π.

So, total flux linked to cube, $\phi_{cube} = \frac{q_{inside}}{\epsilon_0} = \frac{q}{\epsilon_0}$.

Gauss's Law

"According to Gauss Law, the total flux emerging out from a closed surface is equal to $\left(\frac{1}{\epsilon_0}\right)$ times the total charge enclosed by the surface."



Let us consider that there are four charged particles q_1, q_2, q_3, and q_4 in space. We draw a closed gaussian surface which encloses the charges q_1, q_2 and q_3 while the charge q_4 lies outside this surface.

The flux due to charge q_1 passing through the Gaussian surface will be,

$$\oint \vec{E}_1 \cdot d\vec{s} = \frac{q_1}{\epsilon_0} \qquad ...(1)$$

where E_1 = Electric field due to charge q_1 .

Similarly,

$$\oint \vec{E}_2 \cdot d\vec{s} = \frac{q_2}{\epsilon_0} \qquad ...(2)$$

$$\oint \vec{E}_3 \cdot d\vec{s} = \frac{q_3}{\epsilon_0} \qquad ...(3)$$

$$\oint \vec{E}_4 \cdot d\vec{s} = 0 \qquad ...(4)$$

(Since q_4 is outside the Gaussian Surface)

Now, on adding equations (1) + (2) + (3) + (4), we get,

$$\oint (\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4) \cdot d\vec{s} = \frac{(q_1 + q_2 + q_3)}{\epsilon_0} \quad \Rightarrow \oint \vec{E}_{net} \cdot d\vec{s} = \frac{q_{enclosed}}{\epsilon_0}$$

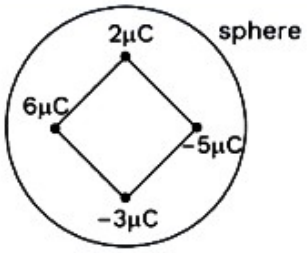
Circle on the integral sign represents that the gaussian surface assumed is a closed surface. $\vec{E}_{net}$ is the electric field due to all the charges whether inside or outside the gaussian surface.

$q_{enclosed}$ is the total charge enclosed within the gaussian surface only.

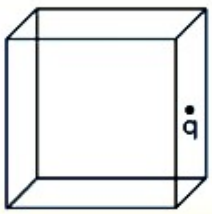
Q. **A square of side 20 cm is enclosed by a spherical surface of radius 80 cm. Square and sphere have the same centre. Four charges +2 × 10^{-6}C, −5 × 10^{-6}C, −3 × 10^{-6}C, + 6 × 10^{-6}C are located at the four corners of a square, then outgoing flux from spherical surface in N-m^2/C will be**

Sol: Flux linked to the spherical surface,

$$\phi_S = \frac{q_{in}}{\epsilon_0} = \frac{2\mu C - 5\mu C - 3\mu C + 6\mu C}{\varepsilon_0} = 0$$



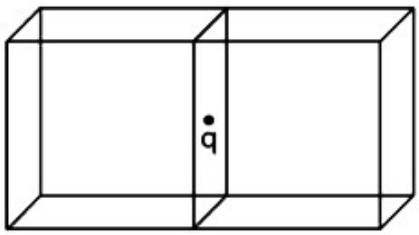
Q. **A charge q is placed at the centre of one face of cube of edge 'a'. Find the electric flux linked to the cube.**



Sol: We can draw another cube in such a way that the charge q is completely enclosed by the two cubes as shown.

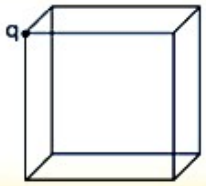
So, the electric flux linked to the two cubes,

$$\phi_{\text{two cubes}} = 2\phi_{\text{cube}} = \frac{q_{in}}{\epsilon_0} = \frac{q}{\epsilon_0}$$



Then, flux linked to single cube, $\phi_{\text{cube}} = \frac{q}{2\,\epsilon_0}$

Q. **A point charge q is placed at a corner of a cube of edge length a. Find total flux coming out from the cube.**

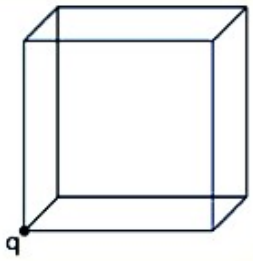


Sol: Charge q can be enclosed completely by 8 cubes placed symmetrically, therefore

Flux linked to 8 cubes, $\phi_{8\text{cubes}} = 8\phi_{\text{cube}} = \frac{q_{in}}{\epsilon_0} = \frac{q}{\epsilon_0}$

Electric flux linked to 1 cube, $\phi_{\text{cube}} = \frac{q}{8\,\epsilon_0}$

Q. **A point charge q is placed at the corner of a cube of edge a. Find the electric flux coming out from a face which is not adjoining the corner where charge is placed.**

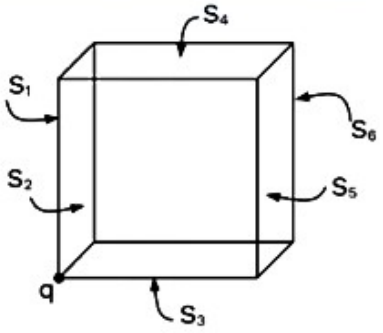


Sol: Surfaces S_1(left), S_2(front), S_3(bottom) are identical w.r.t., q therefore flux passing through them are equal.

$$\phi_1 = \phi_2 = \phi_3$$

Similarly surfaces S_4, S_5 and S_6 are also identical w.r.t., q, therefore, flux passing through them are equal.

$$\phi_4 = \phi_5 = \phi_6$$



Also, at every point on S_1, S_2 and S_3, $\vec{E}$ is perpendicular to $d\vec{A}$.
Therefore,

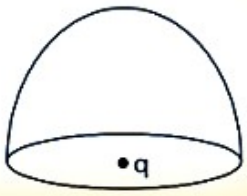
$$\phi_1 = \oint_{1/2/3} \vec{E}.d\vec{A} = 0$$

or $$\phi_1 = \phi_2 = \phi_3 = 0$$

$\because$ $$\phi_{cube} = 3\phi_1 + 3\phi_4 = \frac{q}{8\epsilon_0} \Rightarrow 0 + 3\phi_4 = \frac{q}{8\epsilon_0}$$

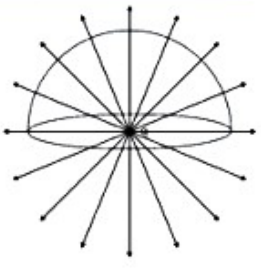
$\therefore$ $$\phi_4 = \phi_5 = \phi_6 = \frac{q}{24\epsilon_0}$$

Q. **Determine the electric flux of a point charge 'q' through the curved hemispherical surface when charge is kept at its centre.**



Sol: It is clear from the figure that half of the flux due to the point charge is passing through the curved surface.

Therefore, $\phi_{hemisphere} = \frac{q}{2\epsilon_0}$



Q. **The inward and outward electric flux from a closed surface are 8×10^3 and 4×10^3 units, respectively. Then the net charge inside the closed surface is?**

Sol: Net flux through the surface, $\phi_{surface} = \phi_{inward} + \phi_{outward} = \left(-8 \times 10^3\right) + \left(4 \times 10^3\right)$

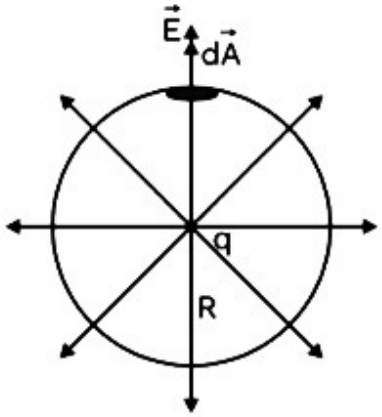
$$\phi_{surface} = -4 \times 10^3 = \frac{q_{in}}{\epsilon_0}$$

Charge enclosed in the surface, $q_{in} = -\left(4 \times 10^3\right) \epsilon_0$ units

Finding electric field due to various charge distributions using Gauss's law

(1) Electric field due to point charge

To find the electric field due to a point charge, draw an imaginary gaussian surface which is symmetric and spherical in shape enclosing the point charge within it.
Due to symmetry, the electric field will be perpendicular to each point on the hypothetical Gaussian surface or field will be parallel to the area vector. Also, due to symmetry, $|\vec{E}|$ will be same at each point.
From Gauss's law,



$$\phi = \oint \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\oint E dA \cos 0° = \int E dA = E \int dA = \frac{q}{\epsilon_0}$$

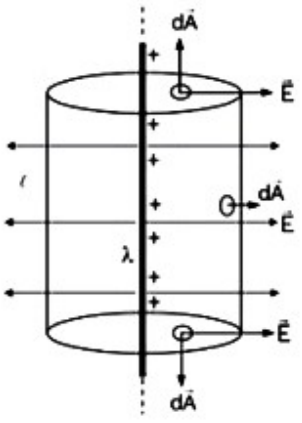
$$E\left(4\pi r^2\right) = \frac{q}{\epsilon_0} \Rightarrow E = \frac{q}{4\pi \epsilon_0 r^2}$$

or $$\vec{E} = \frac{q}{4\pi \epsilon_0 r^2} \hat{r}$$

Here, we can choose a closed surface other than sphere, but we can't solve $\oint \vec{E}.d\vec{A}$ because of the different values of θ and E at each point.

(2) Electric field due uniformly charged infinite wire

We can draw a cylindrical gaussian surface around a uniformly charged wire of infinite length.
Due to symmetry reasons, we choose a hypothetical gaussian cylinder of length ℓ and radius R with its axis same as the line charge.



Now, $\phi_{cylinder} = \phi_{curved} + \phi_{top} + \phi_{bottom}$

$$\oint_{cylinder} \vec{E}.d\vec{A} = \left[\int_{curved} \vec{E}.d\vec{A} + 2\int_{flat} \vec{E}.d\vec{A}\right] = \frac{q_{in}}{\epsilon_0}$$

$$\int E.dA\cos 0^{\circ} + 2\int E.dA\cos 90^{\circ} = \frac{\lambda \ell}{\epsilon_0}$$

$$E\int dA + 0 = \frac{\lambda \ell}{\epsilon_0} \Rightarrow E\left(2\pi r\,\ell\right) = \frac{\lambda \ell}{\epsilon_0}$$

So, the electric field at radial distance r from the infinite wire becomes,

$$E = \frac{\lambda}{2\pi\,\epsilon_0\,r}$$

(Radially outwards & away from the axis).

(3) Electric field due to a non-conducting charged sheet of charge density σ

Due to symmetry reasons, we consider a Gaussian cylinder with its axis parallel to the electric field due to the charged sheet.

Now,

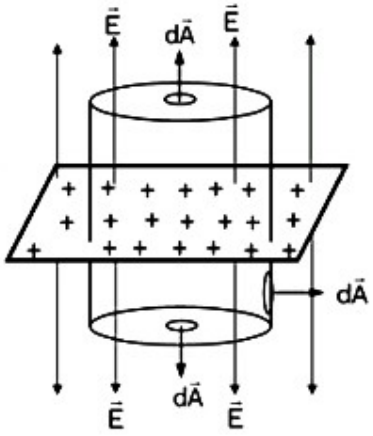
$$\phi_{cylinder} = \phi_{curved} + \phi_{top} + \phi_{bottom} = \frac{q_{in}}{\epsilon_0}$$

$$\oint_{cylinder} \vec{E}.d\vec{A} = \int_{curved} EdA\cos 90^{\circ} + 2\left[\int_{flat} EdA\cos 0^{\circ}\right] = \frac{\sigma\left(\pi R^2\right)}{\epsilon_0}$$

$$0 + 2E\int dA = \sigma\frac{\left(\pi R^2\right)}{\epsilon_0}$$

$$2E\left(\pi R^2\right) = \frac{\sigma\left(\pi R^2\right)}{\epsilon_0}$$

$$E = \frac{\sigma}{2\,\epsilon_0}$$



(4) Electric field due to uniformly charged spherical shell

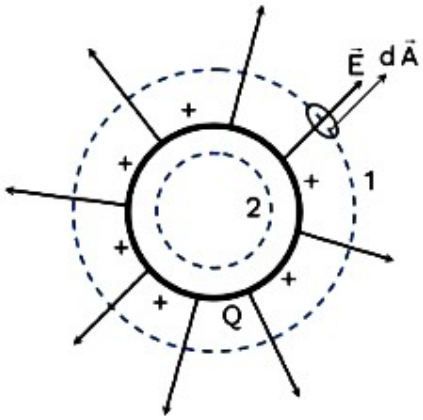
(a) Outside the shell (r ≥ R) :

Consider a Gaussian surface 1 of radius r (>R),

$$\oint \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

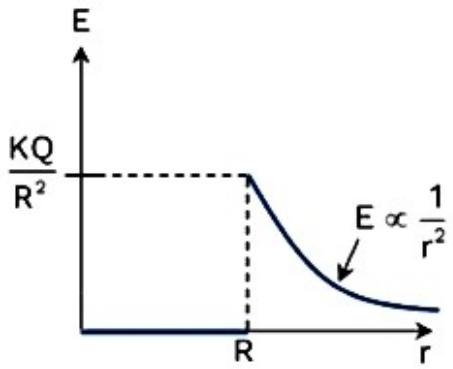
$$\oint E\,dA\cos 0^{\circ} = \frac{Q}{\epsilon_0}$$

$$E\left(4\pi r^2\right) = \frac{Q}{\epsilon_0} \Rightarrow \vec{E} = \left(\frac{Q}{4\pi\,\epsilon_0\,r^2}\right)\hat{r}$$



(b) Inside the shell (r < R):

Consider a Gaussian surface 2 of radius r (<R),



$$\oint \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0} = 0$$

$$\oint E\,dA\cos 0° = 0$$

$$\Rightarrow \quad E\oint dA = 0 \Rightarrow E\left(4\pi r^2\right) = 0$$

$$E = 0.$$

Conclusion:

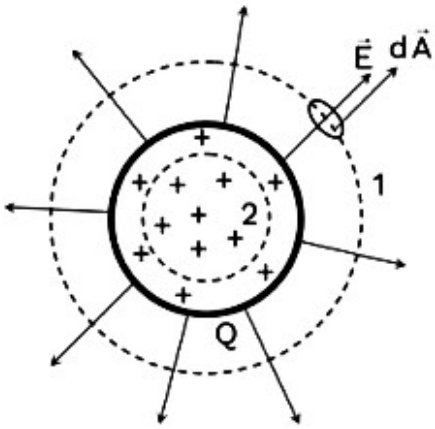
Electric field due to a charged spherical shell is given by,

$$E = \begin{cases} 0 & ; \text{for } r < R. \\ \dfrac{KQ}{r^2} & ; \text{for } r \geq R. \end{cases}$$

(5) Electric field due to a uniformly charged solid non conducting sphere

(a) Outside the sphere (r ≥ R):

For Gaussian surface 1,



$$\oint \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

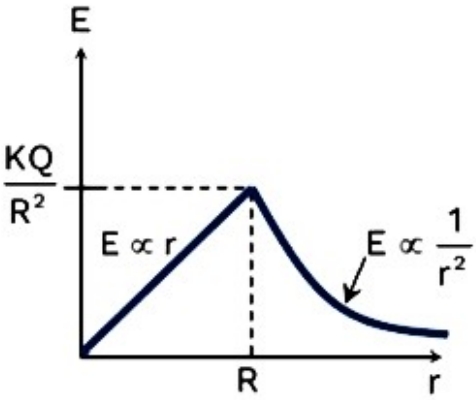
$$\oint E\,dA\cos 0° = \frac{Q}{\epsilon_0}$$

$$E\oint dA = E\left(4\pi r^2\right) = \frac{Q}{\epsilon_0}$$

$$\vec{E} = \frac{Q}{4\pi\,\epsilon_0\,r^2}\hat{r}$$

(b) Inside the sphere (r < R):

For Gaussian surface 2,



$$\oint \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\Rightarrow \quad \oint E\,dA\cos 0° = \frac{Q}{\frac{4\pi}{3}R^3}.\frac{\frac{4\pi}{3}r^3}{\epsilon_0}$$

$$E\oint dA = \frac{Qr^3}{\epsilon_0\,R^3}$$

$$E\left(4\pi r^2\right) = \frac{Qr^3}{\epsilon_0\,R^3} \Rightarrow \vec{E} = \frac{Q}{4\pi\,\epsilon_0\,R^3}\vec{r}$$

Conclusion:

$$E = \begin{cases} \dfrac{KQr}{R^3} & ; \quad \text{for } 0 \le r \le R \\ \dfrac{KQ}{r^2} & ; \quad \text{for } r > R \end{cases}$$

(6) Electric field due to a uniformly charged infinite cylinder

(a) Outside the cylinder (r > R):

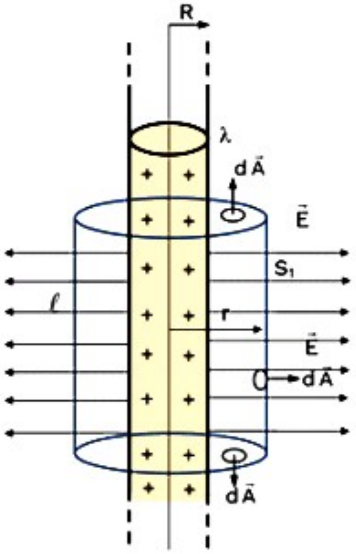
Assume λ be the charge per unit length on cylinder.

Apply Gauss's Law for S_1,

$$\oint_{S_1} \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\int_{\text{curved}} \vec{E}.d\vec{A} + \int_{\text{top}} \vec{E}.d\vec{A} + \int_{\text{bottom}} \vec{E}.d\vec{A} = \frac{\lambda \ell}{\epsilon_0}$$

$$\Rightarrow \quad E = \frac{2K\lambda}{r}$$



(b) Inside the cylinder (r < R):

Effective charge per unit length for S_2,

$$\lambda' = \frac{\lambda}{\pi R^2}.\pi r^2 = \frac{\lambda r^2}{R^2}$$

Apply Gauss's Law for S_2,

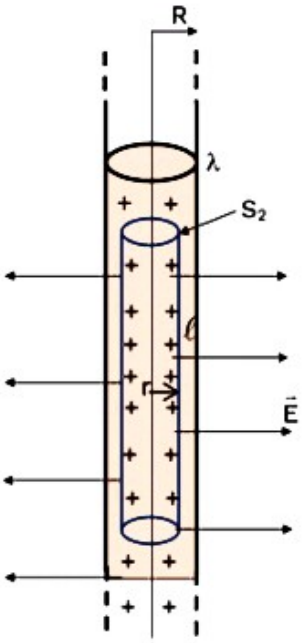
$$\oint \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\int_{\text{curved}} \vec{E}.d\vec{A} + \int_{\text{top}} \vec{E}.d\vec{A} + \int_{\text{bottom}} \vec{E}.d\vec{A} = \frac{\lambda' \ell}{\epsilon_0}$$

$$\int E dA \cos 0^\circ + 0 + 0 = \frac{\lambda r^2 \ell}{R^2 \epsilon_0}$$

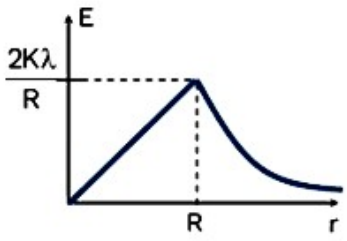
$$E\int dA = \frac{\lambda r^2 \ell}{R^2 \epsilon_0} \quad \Rightarrow \quad E(2\pi r \ell) = \frac{\lambda r^2 \ell}{R^2 \epsilon_0}$$

$$E = \frac{\lambda r}{2\pi \epsilon_0 R^2} \quad \Rightarrow \quad E = \frac{2K\lambda r}{R^2}$$



Conclusion:

$$E = \begin{cases} \dfrac{2K\lambda r}{R^2} & ; \text{ for } 0 \le r \le R. \\ \dfrac{2K\lambda}{r} & ; \text{ for } r > R. \end{cases}$$



(7) Electric field due to uniformly charged cylindrical shell

(a) Outside the cylindrical shell (r > R):

Assume λ be the charge per unit length on cylinder.

Apply Gauss's Law for S_1,

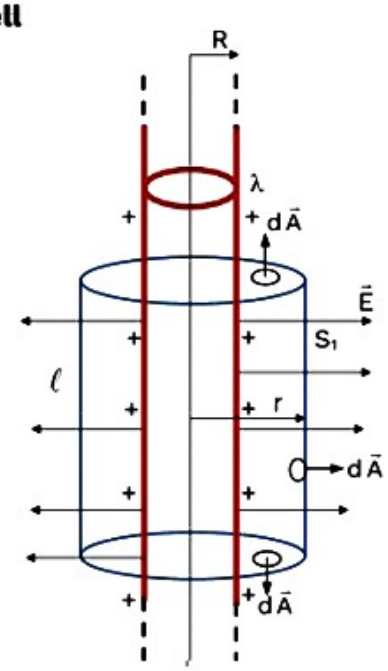
$$\oint_{S_1} \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$\int_{\text{curved}} \vec{E}.d\vec{A} + \int_{\text{top}} \vec{E}.d\vec{A} + \int_{\text{bottom}} \vec{E}.d\vec{A} = \frac{\lambda \ell}{\epsilon_0}$$

$$\int E\,dA + 0 + 0 = \frac{\lambda \ell}{\epsilon_0}$$

$$E(2\pi r\ell) = \frac{\lambda \ell}{\epsilon_0} \Rightarrow E = \frac{\lambda}{2\pi \epsilon_0 r}$$

$$E = \frac{2K\lambda}{r}$$



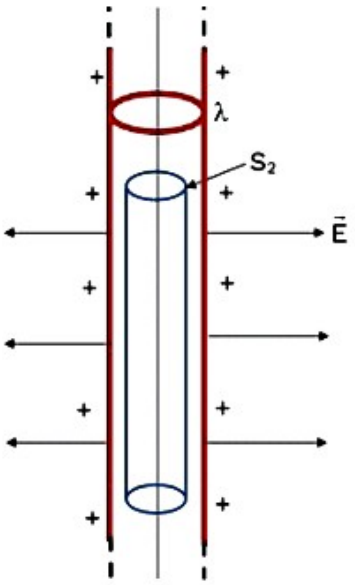
(b) Inside the cylindrical shell (r < R):

Apply Gauss's law on S_2,

$$\oint_{S_2} \vec{E}.d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

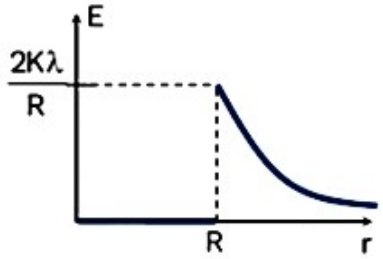
$$\Rightarrow \quad \oint \vec{E}.d\vec{A} = 0$$

$$E = 0$$



Conclusion:

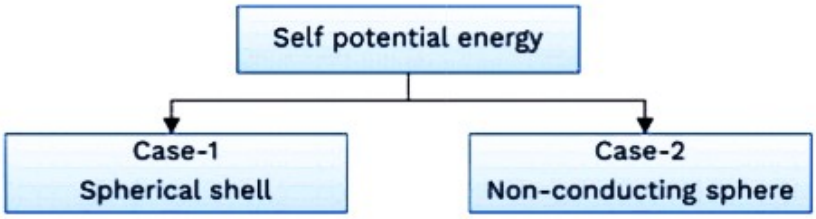
$$E = \begin{cases} 0 & ; \quad \text{for } r < R. \\ \dfrac{2K\lambda}{r} & ; \quad \text{for } r \geq R. \end{cases}$$



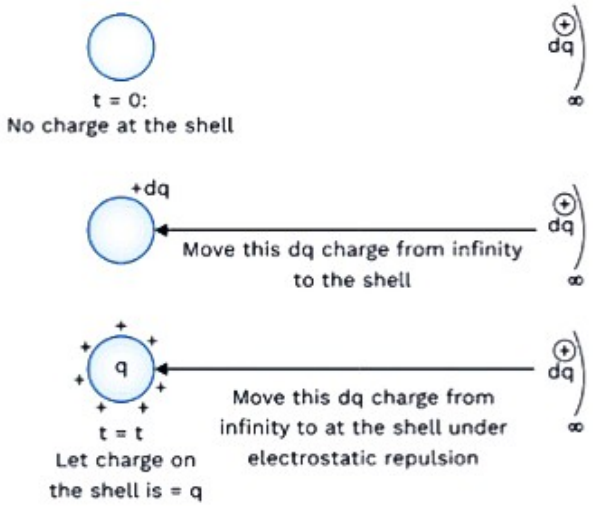
Self-Potential Energy

The electrostatic energy stored in a body due to the charges present on it is called its self- potential energy. It is equal to the total work done in bringing the total charge on the body very very slowly from infinity.

We will derive the result for the self-potential energy for two bodies here.



Case-1 : Self potential energy of a charged spherical shell



Change in potential energy of shell is equal to work done by external force in moving the dq charge from infinity to the shell.

This work done by external force is stored in the form of potential energy of shell called self-potential energy of the shell.

$$dW_{ext} = dU$$

$$dW = dq\,(dV)$$

$$dW = dq\,(V_R - V_\infty)$$

$$dW = dq\left(\frac{Kq}{R} - 0\right)$$

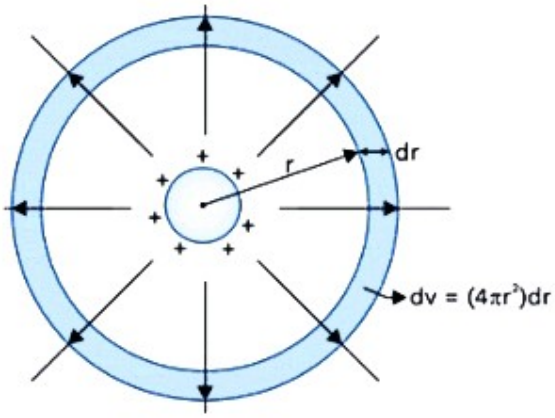
$$W = \int dW = \int_0^Q \frac{Kq}{R}\,dq$$

$$= \frac{K}{R}\int_0^Q q\,dq = \frac{K}{R}\left(\frac{q^2}{2}\right)_0^Q$$

$$\boxed{\text{Self P.E. of shell} = \frac{KQ^2}{2R}}.$$

Note:
This energy is stored in external electric field of the shell.

Alternate method:



Alternately, we can use the concept of energy density to find the self-potential energy of the shell.

Energy density is given as, $U_D = \frac{1}{2}\epsilon_0 E^2$

Since we know that this energy is stored in the form of electric field which is confined in the region between R to ∞, so
Energy stored in dV volume,
dU = (Energy density) (dV)

$$U = \int dU = \int_R^\infty \left(\frac{1}{2}\epsilon_0 E^2\right) \times \left(4\pi r^2 dr\right)$$

$$U = \frac{1}{2}\epsilon_0 \times 4\pi \int_R^\infty \left(\frac{KQ}{r^2}\right)^2 r^2\, dr$$

$$= \frac{4\pi\epsilon_0}{2} \times K^2 \times Q^2 \times \int_R^\infty \frac{1}{r^2}\, dr$$

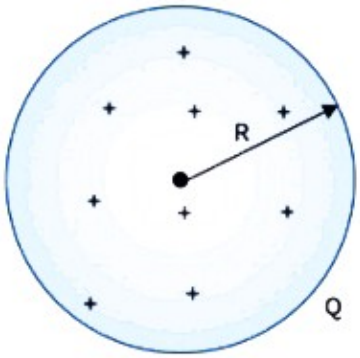
$$= \frac{K^2}{2K} \times Q^2 \times \left(-\frac{1}{r}\right)_R^\infty$$

$$= -\frac{K}{2} \times Q^2 \left[\frac{1}{\infty} - \frac{1}{R}\right]$$

$$= -\frac{KQ^2}{2}\left[0 - \frac{1}{R}\right]$$

$$\boxed{U = -\frac{KQ^2}{2R}}.$$

Case 2: Self-Potential Energy of solid non-conducing sphere



Energy Density = $\frac{1}{2}\epsilon_0 E^2$.

In this case energy is stored inside as well as outside the sphere in the form of electric field.

$$\int dU = \int_0^\infty \frac{1}{2}\epsilon_0 E^2 dV$$

$$dU = \underbrace{\int_0^R \frac{1}{2}\epsilon_0 E^2\, dV}_{\text{inside}} + \underbrace{\int_R^\infty \left(\frac{1}{2}\epsilon_0 E^2\right)\cdot dV}_{\text{outside}}$$

$$= \int_0^R \frac{1}{2}\ \epsilon_0 \left[\frac{KQ(r)}{R^3}\right]^2 \left(4\pi r^2\right) dr + \int_R^\infty \frac{1}{2}\epsilon_0 \left(\frac{KQ}{r^2}\right)^2 \left(4\pi r^2 dr\right)$$

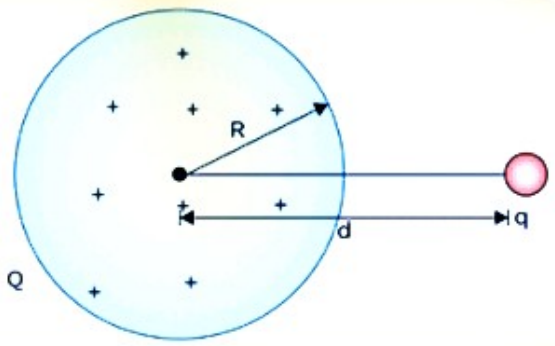
$$U = \frac{KQ^2}{10R} + \frac{KQ^2}{2R}$$

$$U = \frac{KQ^2 + 5KQ^2}{10R} = \frac{6KQ^2}{10R}$$

$$\boxed{U = \frac{3}{5}\frac{KQ^2}{R}}$$

$U_{\text{Total P.E.}}$	=	U_{inside}	+	U_{outside}
$\frac{3}{5}\frac{kQ^2}{R}$	=	$\frac{kQ^2}{10R}$	+	$\frac{kQ^2}{2R}$
↓		↓		↓
Total self potential energy of sphere		Energy stored in the form of internal $\vec{E}$-field		Energy stored in the form of external $\vec{E}$-field

Q. **Find out the electrostatic potential energy of the system shown in the figure below.**



Sol: Total potential energy of system = Self P.E. of non-conducting sphere + Interaction potential energy

$$U_T = \frac{3}{5}\frac{KQ^2}{R} + \frac{KQq}{d}$$

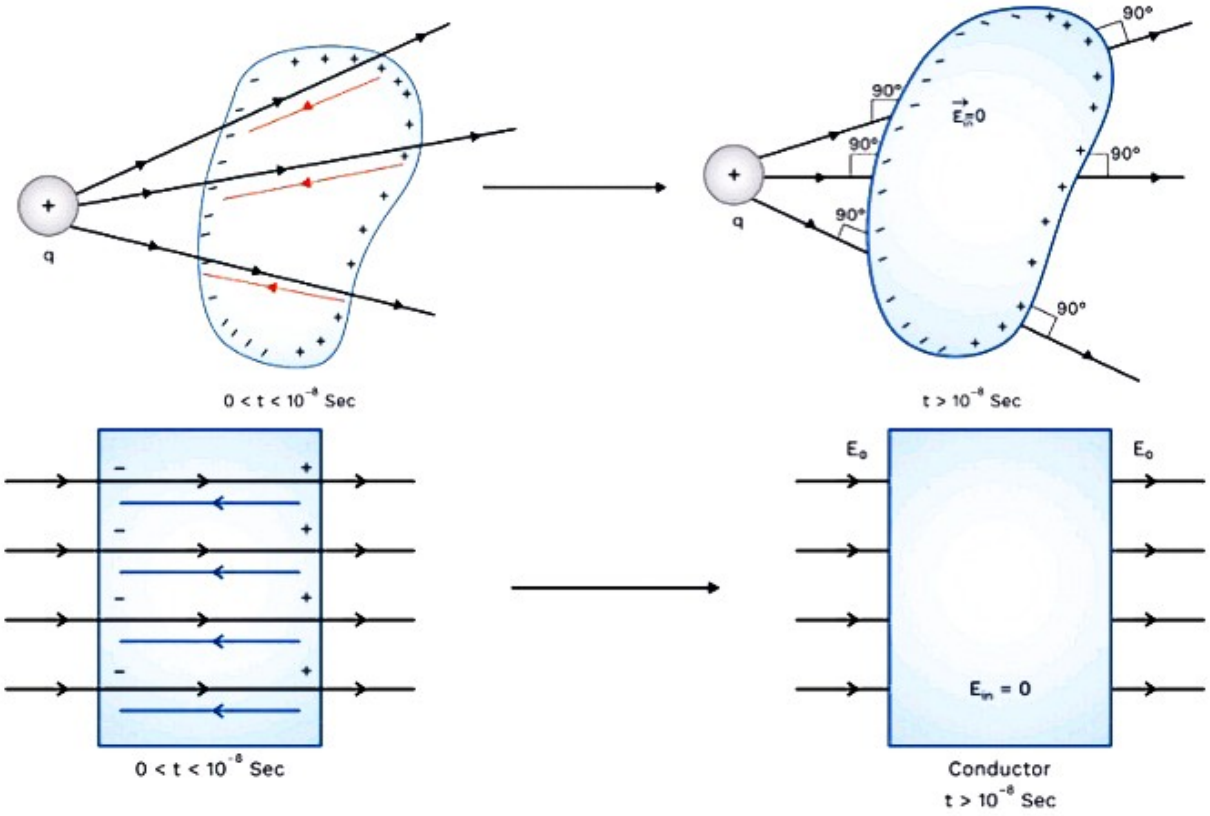
Electrostatics of Conductors

Conductors

The substances having large number of free electrons are known as conductors. It is a type of material which allows the flow of charges.

Redistribution of charges in conductors

When a charge is placed near a conductor, there is redistribution of the charges in the conductor. On placing a positive charge near a conductor, the negative charges (free electrons) accumulate on the surface which is closer to the external positive charge. The vacancies created due to the free electrons after moving act as positive charges on the conductor.



For all practical purposes, the electric field inside the body of a conductor is zero.

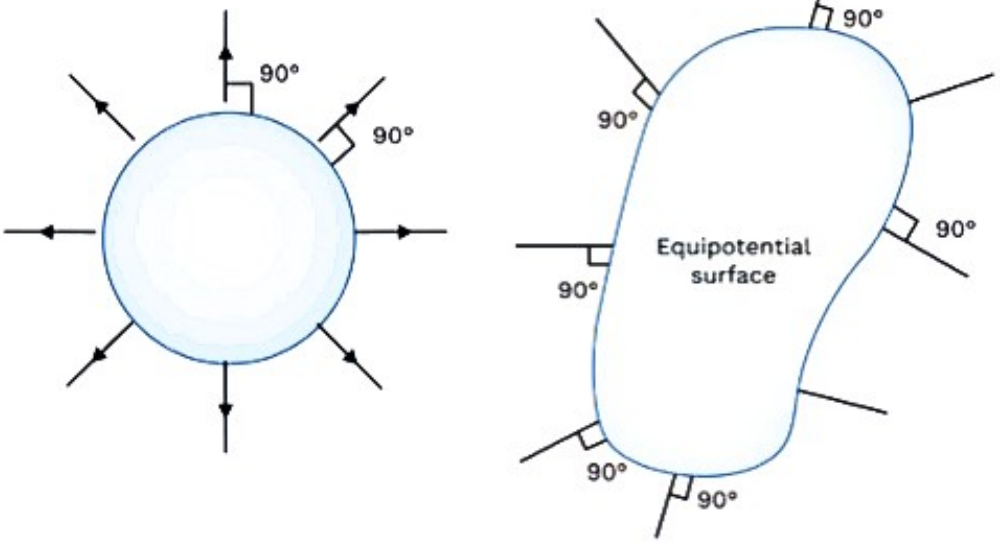
When a conductor is placed in an external electric field, there will be redistribution of charges such that the net electric field inside the material of the conductor becomes zero under the 'electrostatic condition'.

Properties of conductor

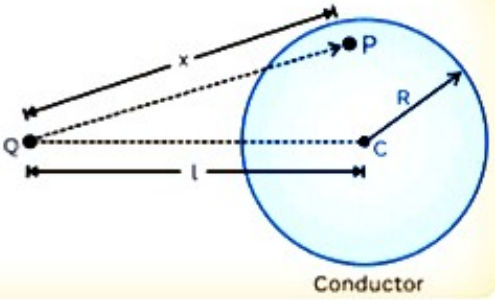
(1) There is no electric field inside the conductor either the conductor is solid or hollow.

(2) Any charge given to the isolated conductor always resides at the surface of the conductor.

(3) The surface of the conductor is equipotential, hence electric field lines are always normal to the surface of the conductor.



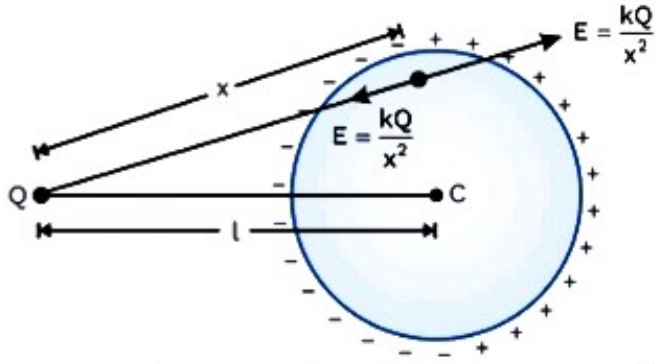
Q. **(a) Calculate the electric field and electric potential at point 'P' in the given figure.**
(b) Calculate the electric field and electric potential at point 'P' due to induced charges.



Sol: (a) Since it is a conductor so there is no electric field inside the conductor.

So, the electric field at point P due to the point charge Q must be balanced by the electric field due to the induced charges on the surface of the conductor.

$$\boxed{E_p = 0}$$



Since it is a conductor, and a conductor is an equipotential surface, so potential at point P will be equal to the potential at the centre C.

$$V_P = V_C = \frac{kq}{\ell} + \text{ Potential due to induced charges.}$$

$$V_P = \frac{kq}{\ell} + 0 \text{ (due to symmetry)}$$

$$\boxed{V_P = \frac{kq}{\ell}}$$

(b) Electric field at P due to induced charges, $E_{induced} = \frac{KQ}{x^2}$

[Towards Q, just to compensate the field produced by Q].

Also,

$V_C = V_P$

$\frac{kQ}{\ell} = \frac{kQ}{x}$ + Potential due to induced charges.

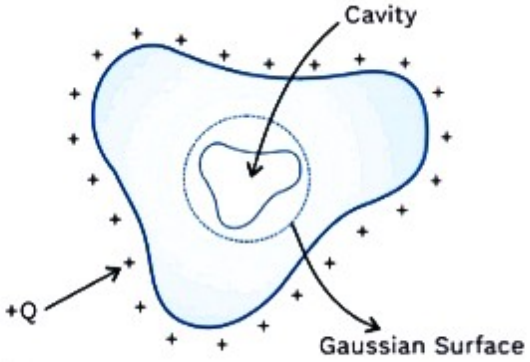
$\frac{kQ}{\ell} = \frac{kQ}{x} + V_{ind}$

$V_{ind} = \frac{kQ}{\ell} - \frac{kQ}{x}$

$$\boxed{V_{\text{due to induced charge}} = kQ\left[\frac{1}{\ell} - \frac{1}{x}\right]}$$

Concept of cavity inside a conductor

(a) When there is a cavity inside a charged conductor



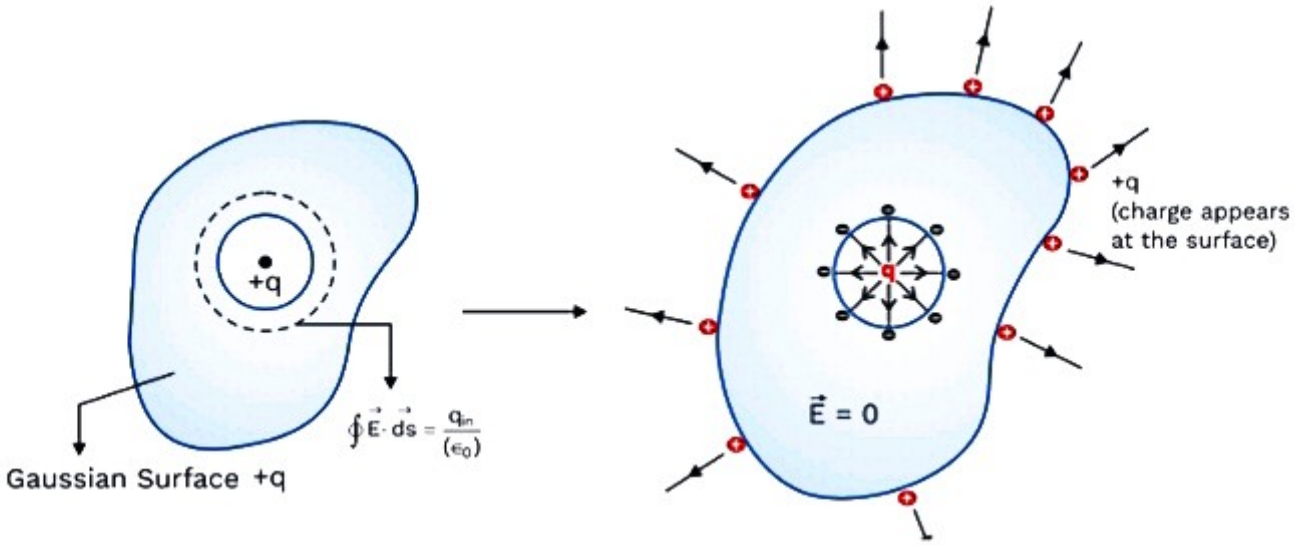
According to Gauss Law,

$$\oint \vec{E} \cdot \overrightarrow{ds} = \frac{q_{in}}{\epsilon_0}$$

$$\oint (0) \cdot \overrightarrow{ds} = \frac{q_{in}}{\epsilon_0}$$

$\boxed{q_{in} = 0}$ → No charge is induced inside the cavity.

(b) When a charge is present inside the cavity

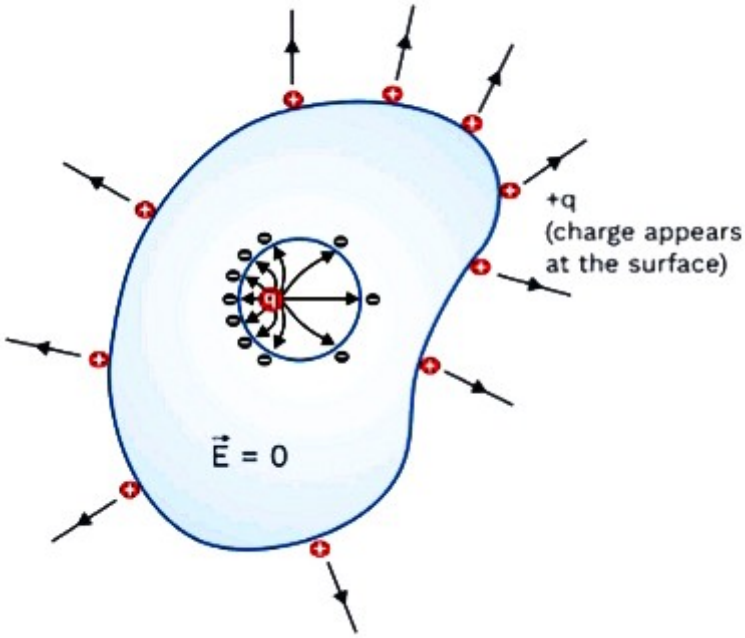


$$\oint \vec{E} \cdot \overrightarrow{ds} = \frac{q_{in}}{\epsilon_0}$$

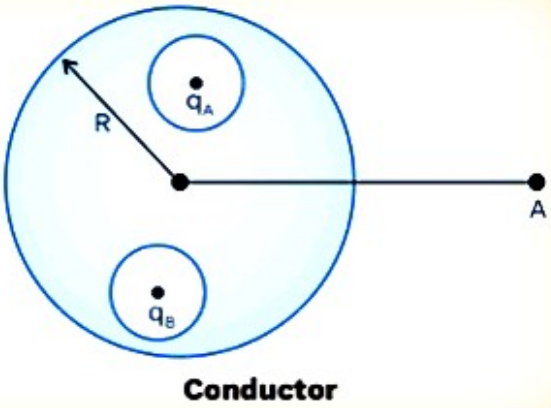
$$\oint (0) \cdot \overrightarrow{ds} = \frac{q_{in}}{\epsilon_0} = q + q_{induced}$$

Induced charge ($q_{induced}$) at inner surface of cavity = −q.

On displacing the positive charge inside the cavity, there is no change in the charge distribution at the outer surface of the conductor, but the arrangement of the induced charge inside the cavity changes.

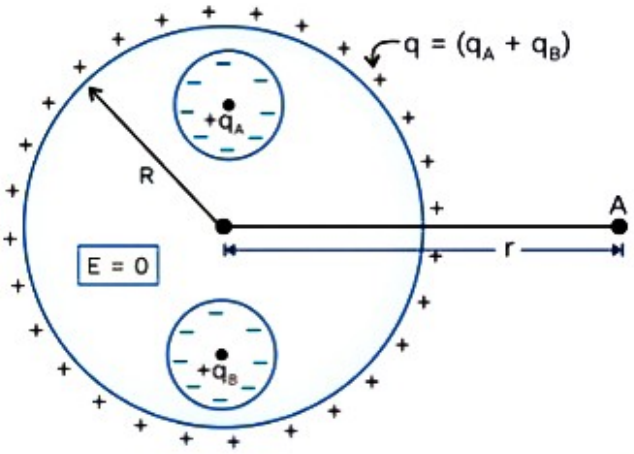


Q. **Calculate the electric potential and electric field at point 'A' in the given figure.**

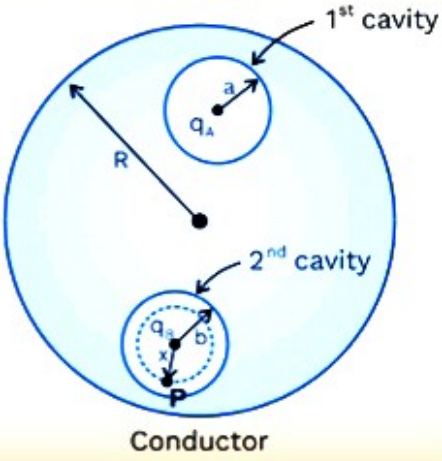


Sol: Electric field $E_A = \frac{K(q_A + q_B)}{r^2}$

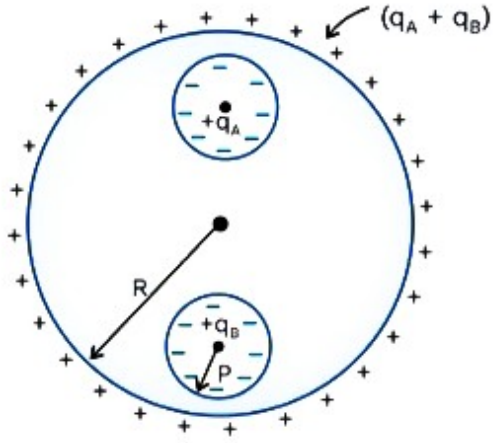
Electric potential $V_A = \frac{K(q_A + q_B)}{r}$



Q. **Find electric field and potential at point 'P' which is inside the sphere of radius (R).**



Sol:



$$E_p = \frac{kq_B}{x^2} + 0 + 0 + 0$$

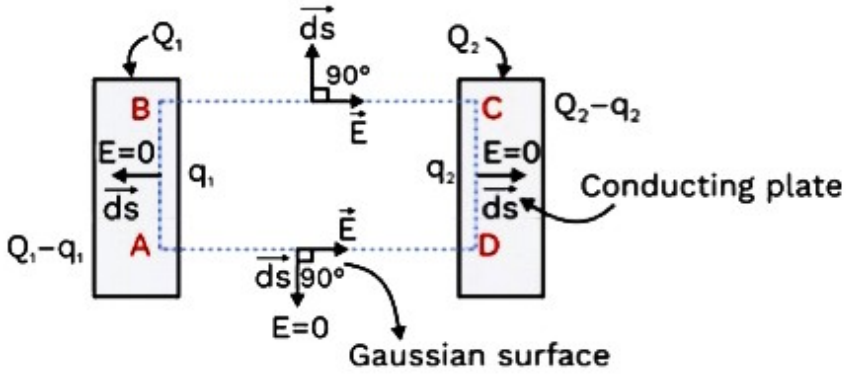
- $\frac{kq_B}{x^2}$: Due to first cavity
- 0: Due to charge induced inside 2nd cavity
- 0: Outer surface charge of sphere

Electric field at point 'P', $E_p = \frac{K q_B}{x^2}$

Electric potential at point 'P' , $V_p = \frac{K q_B}{x} + \frac{K(-q_B)}{b} + \frac{K(q_A + q_B)}{R}$

Charge distribution on conducting parallel plates

Suppose there are two parallel plate conductors shown in figure. Now charge Q_1 is given to the left plate and charge Q_2 is given to the right plate.



According to Gauss law :

$$\oint_{ABCD} \vec{E}\cdot\overrightarrow{ds} = \frac{q_{in}}{\epsilon_0}$$

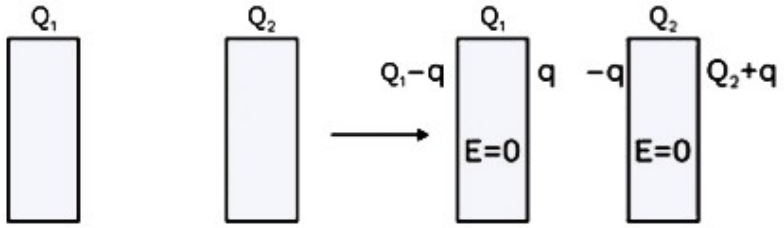
$$\oint_{AB} \vec{E}\cdot\overrightarrow{ds} + \int_{BC} \vec{E}\cdot\overrightarrow{ds} + \int_{CD} \vec{E}\cdot\overrightarrow{ds} + \int_{DA} \vec{E}\cdot\overrightarrow{ds} = \left(\frac{q_1 + q_2}{\epsilon_0}\right)$$

$$\int_{AB} (0)\cdot\overrightarrow{ds} + \int_{BC} |\vec{E}||\overrightarrow{ds}|\cos 90° + \int_{CD} (0)\cdot\overrightarrow{ds} + \int_{DA} |\vec{E}||\overrightarrow{ds}|\cos 90° = \left(\frac{q_1 + q_2}{\epsilon_0}\right)$$

$$0 = \left(\frac{q_1 + q_2}{\epsilon_0}\right) \Rightarrow \quad q_1 = -q_2$$

Let $|q_1| = |q_2| = q \quad \Rightarrow \quad \boxed{|q_1| = |-q_2| = q}$.

From above result, we can conclude that in parallel plates, inner surface always get equal and opposite charges for validity of Gauss law and electric field inside the conductor is zero.



Calculation for value of 'q':

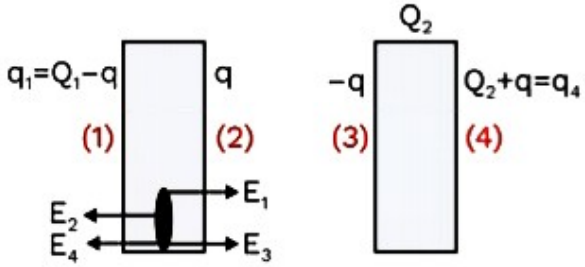


Plate area = A

Since net electric field inside the conductor is zero,

$$\vec{E}_{net} = 0$$

$$\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 = 0 \quad \Rightarrow \quad E_1 + E_3 - E_2 - E_4 = 0$$

$|\vec{E}_2| = |\vec{E}_3|$; (Due to equal charges on inner surfaces)

$$E_1 + E_3 = E_2 + E_4$$

$$|E_2| = |E_3|$$

$$E_1 = E_4 \quad \Rightarrow \quad \frac{\sigma_1}{2\epsilon_0} = \frac{\sigma_4}{2\epsilon_0}$$

$$\frac{q_1}{A(2\epsilon_0)} = \frac{q_4}{A(2\epsilon_0)}$$

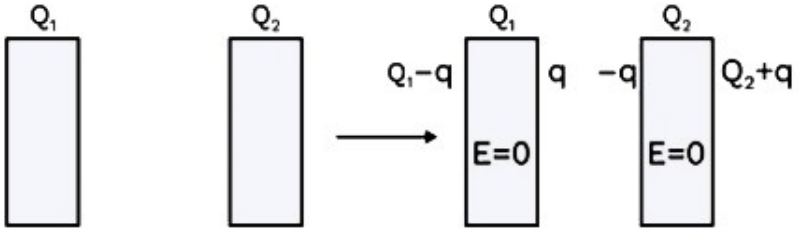
$$(Q_1 - q) = (Q_2 + q)$$

$$Q_1 - Q_2 = 2q$$

$$0 = \left(\frac{q_1 + q_2}{\epsilon_0}\right) \Rightarrow \quad q_1 = -q_2$$

Let $|q_1| = |q_2| = q \quad \Rightarrow \quad \boxed{|q_1| = |-q_2| = q}$.

From above result, we can conclude that in parallel plates, inner surface always get equal and opposite charges for validity of Gauss law and electric field inside the conductor is zero.



Calculation for value of 'q':

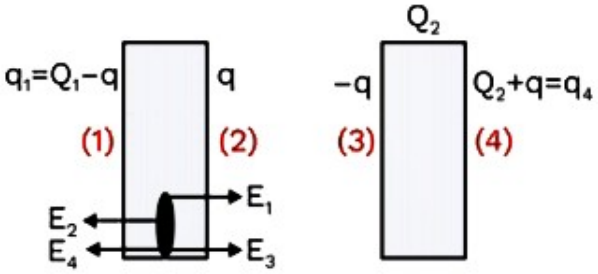


Plate area = A

Since net electric field inside the conductor is zero,

$$\vec{E}_{net} = 0$$

$$\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4 = 0 \quad \Rightarrow \quad E_1 + E_3 - E_2 - E_4 = 0$$

$|\vec{E}_2| = |\vec{E}_3|$; (Due to equal charges on inner surfaces)

$$E_1 + E_3 = E_2 + E_4$$

$$|E_2| = |E_3|$$

$$E_1 = E_4 \quad \Rightarrow \quad \frac{\sigma_1}{2\epsilon_0} = \frac{\sigma_4}{2\epsilon_0}$$

$$\frac{q_1}{A(2\epsilon_0)} = \frac{q_4}{A(2\epsilon_0)}$$

$$(Q_1 - q) = (Q_2 + q)$$

$$Q_1 - Q_2 = 2q$$

$$\boxed{q = \frac{Q_1 - Q_2}{2}}.$$

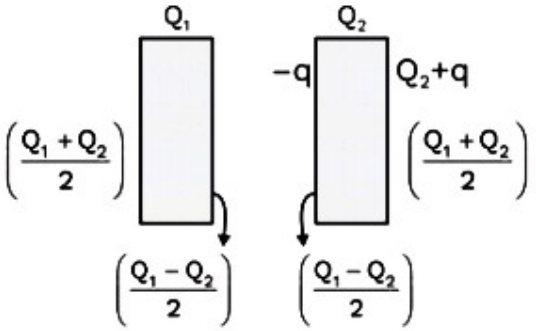
Now charge on outer surface;

$$q_{out} = Q_1 - q = Q_1 - \left(\frac{Q_1 - Q_2}{2}\right)$$

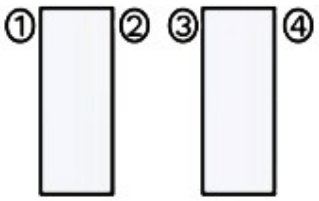
$$q_{out} = \frac{2Q_1 - Q_1 + Q_2}{2}$$

$$\boxed{q_{out} = \frac{Q_1 + Q_2}{2}}.$$

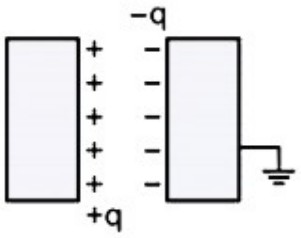
Final charge distribution on plates :



Important points to note-

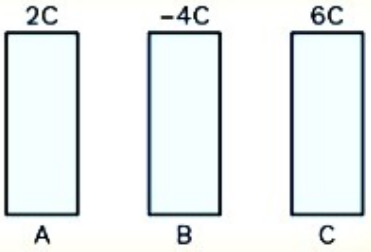
1. Facing surfaces have equal but opposite nature of charges.
2. Outer surface always has equal charges of magnitude equal to half of the summation of charges.
3. 

   The charge appearing on surface (2) and (3) is called bonded charge, and the charge appearing on surface (1) and (4) is called free charge.

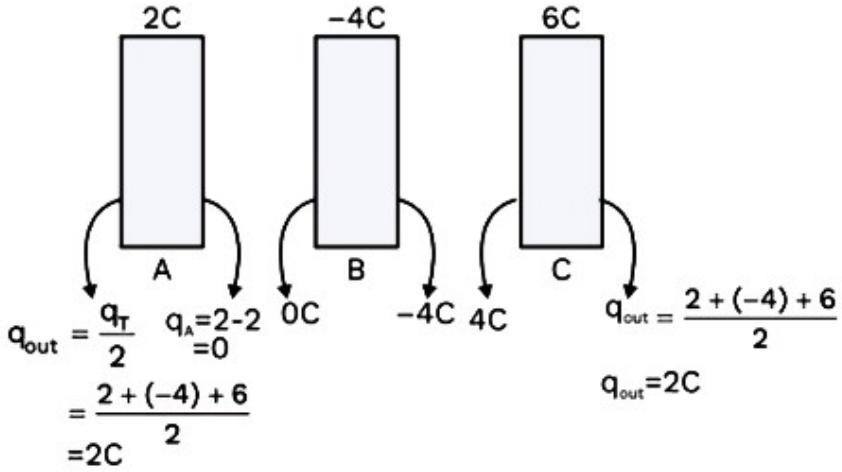
4. 

   If we join the second plate (Right one) with ground, the charge appearing on the surface (4) will go to the earth.

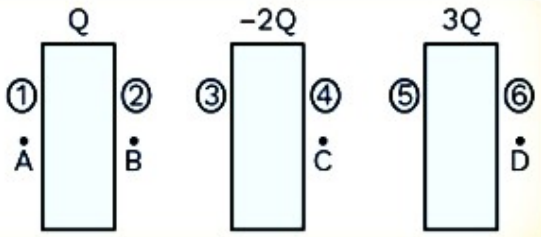
Q. **There are three conducting plates A, B, C. Charges given to each plate are 2C, −4C and 6C respectively. Find the charge on each side of the given plates.**



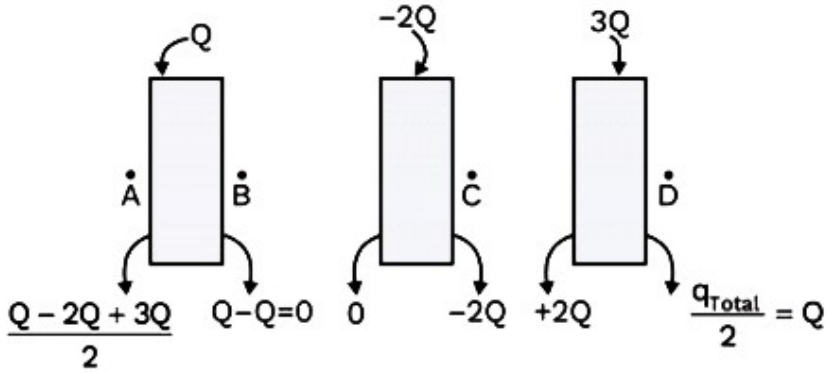
Sol.



Q. **Suppose there are three plates A, B, C having charges Q, -2Q and 3Q respectively. Find out the electric field at points A, B, C, D.**

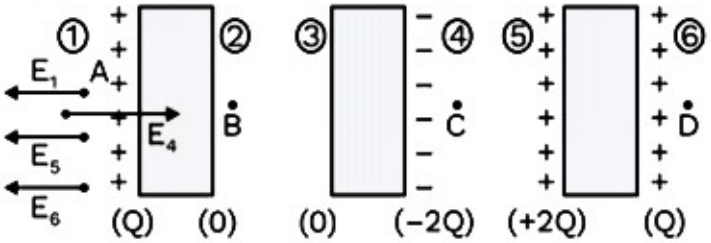


Sol. There are 6 surfaces of given three plates as shown in figure. Distribute the charges on each side of the plates.



Let area of plates is 'A'

(1) Electric field at point 'A'



$E_A = E_1 + E_5 + E_6 - E_4$; (Let left direction be +ve).

$|E_5| = |E_4|$

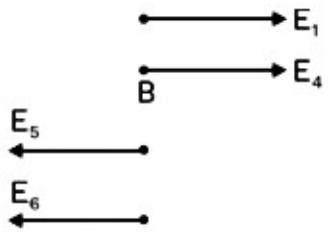
$E_A = E_1 + E_6$

$$= \frac{Q}{2\epsilon_0(A)} + \frac{Q}{2\epsilon_0(A)}$$

$$\boxed{E_A = \frac{Q}{\epsilon_0 A} \text{ (towards left)}}$$

Note : Since there is no charge at surface (2) and (3), so $E_2 = 0$ and $E_3 = 0$

(2) Electric field at point 'B'

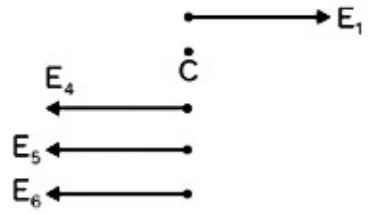


$|\vec{E}_B| = E_1 + E_4 - E_5 - E_6$

$|E_4| = |E_5|$; $|E_1| = |E_6|$

$|\vec{E}_B| = E_1 - E_6 = 0$

(3) Electric field at point 'C'



$E_C = E_4 + E_5 + E_6 - E_1$

$|E_1| = |E_6|$

$E_C = E_4 + E_5$

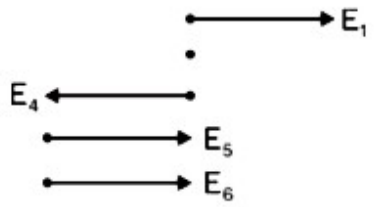
$$= \frac{2Q}{A(2\epsilon_0)} + \frac{2Q}{2(A\epsilon_0)}$$

$$E_C = \frac{2Q}{A\epsilon_0} \text{ (towards left).}$$

(4) Electric field at point 'D'

$E_D = E_1 + E_5 + E_6 - E_4$; (here we take right direction as +ve direction)

$|E_4| = |E_5|$

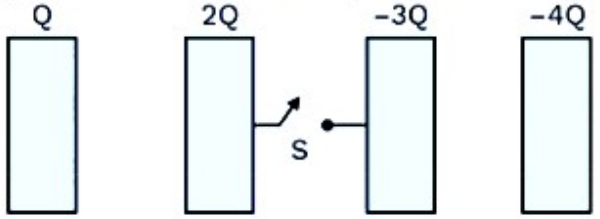


$E_D = E_1 + E_6$

$$= \frac{Q}{A(2\epsilon_0)} + \frac{Q}{A(2\epsilon_0)}$$

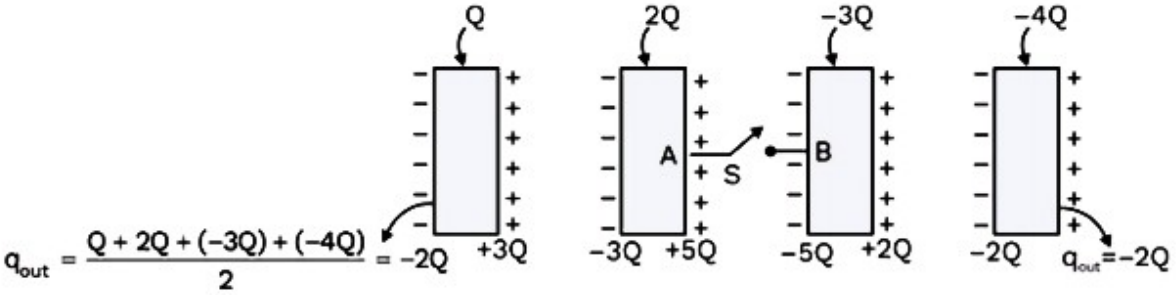
$$E_D = \frac{Q}{A\epsilon_0} \text{ (towards right).}$$

Q. **Large conducting plates are shown in figure.**



If switch 'S' is closed, then find the charge that will flow through Wire.

Sol. **Step-1:** Distribute the charges on each side of the plates.



$$q_{out} = \frac{Q + 2Q + (-3Q) + (-4Q)}{2} = -2Q$$

Now, on connecting with the wire, the charges 5Q and −5Q will get neutralized. Hence 5Q charge will flow for A to B.

Note :

Sum of charges given to each side of plate is equal to charge given to that particular plate.

Sharing of charges

Two conducting hollow spherical shells of radii R_1 and R_2 having charges Q_1 and Q_2 respectively and separated by large distance, are joined by a conducting wire.



Let final charges on spheres are q_1 and q_2 respectively. Potential on both spherical shells become equal after joining, therefore

$$\frac{Kq_1}{R_1} = \frac{Kq_2}{R_2} \Rightarrow \frac{q_1}{q_2} = \frac{R_1}{R_2} \quad \text{... (i)}$$

And, by charge conservation, $q_1 + q_2 = Q_1 + Q_2$
From (i) and (ii) :

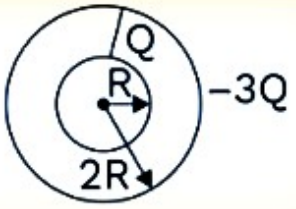
$$q_1 = \frac{(Q_1 + Q_2)R_1}{R_1 + R_2}, \quad q_2 = \frac{(Q_1 + Q_2)R_2}{R_1 + R_2}$$

Ratio of final charges:

$$\frac{q_1}{q_2} = \frac{R_1}{R_2} \Rightarrow \frac{\sigma_1 4\pi R_1^2}{\sigma_2 4\pi R_2^2} = \frac{R_1}{R_2}$$

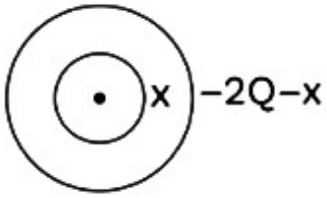
$\therefore$ Ratio of surface charge densities: $\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$

Q. **The two conducting spherical shells are joined by a conducting wire and cut after some time when charge stops flowing. Find out the charge on each sphere after that.**



Sol: After cutting the wire, the potential of both the shells is equal.
Thus, potential of inner shell

$$V_{in} = \frac{Kx}{R} + \frac{K(-2Q - x)}{2R} = \frac{K(x - 2Q)}{2R}$$



And, potential of outer shell

$$V_{out} = \frac{Kx}{2R} + \frac{K(-2Q - x)}{2R} = \frac{-KQ}{R}$$

As, $V_{out} = V_{in}$

$$\Rightarrow \frac{-KQ}{R} = \frac{K(x - 2Q)}{2R} \Rightarrow -2Q = x - 2Q \Rightarrow x = 0$$

So, charge on inner spherical shell = 0 and outer spherical shell = – 2Q.

Q. **An isolated conducting sphere of charge Q and radius R is connected to a similar uncharged sphere (kept at a large distance) by using a high resistance wire. After a long time, what is the amount of heat loss?**

Sol: When two conducting spheres of equal radii are connected, charge is equally distributed on them. So, we can say that heat loss of system:

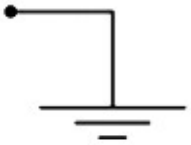
$$\Delta H = U_i - U_f = \left(\frac{Q^2}{8\pi\varepsilon_0 R} - 0\right) - \left(\frac{Q^2/4}{8\pi\varepsilon_0 R} + \frac{Q^2/4}{8\pi\varepsilon_0 R}\right) = \frac{Q^2}{16\pi\varepsilon_0 R}$$

Earthing of a conductor

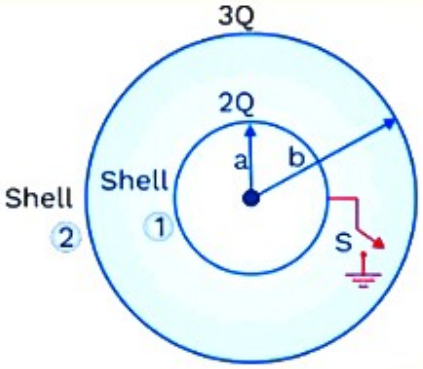
The electric potential of the earth is assumed to be zero. Earth is assumed to be a source of infinte charges and it can supply any amount of charge to a body connected to it to make its potential equal to zero.

Whenever an isolated conductor is grounded [connected to the earth], then charges flow in such a way that the potential of conductor becomes zero.

Symbol used for earthing is as shown below:



Q. **In the following diagram, two concentric shells are shown. When the switch 'S' is closed, then find the new charge appearing on the inner shell.**



Sol:

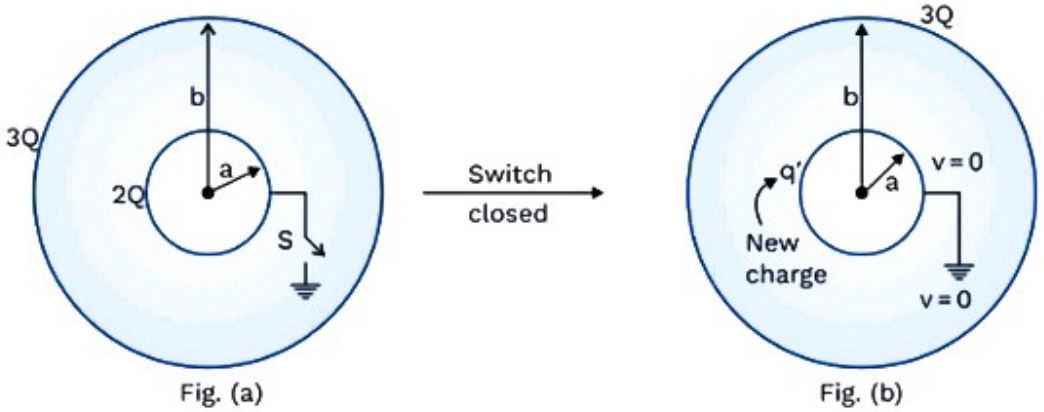


Fig. (a) Fig. (b)

The electric potential of the inner shell,

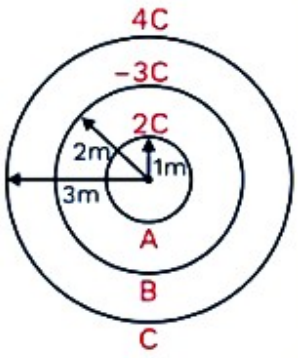
$$V = \frac{Kq'}{a} + \frac{K(3q)}{b}$$

Since, the inner shell is connected to ground so potential of inner shell becomes zero.

$V = 0$

$\frac{Kq'}{a} = -\frac{K(3q)}{b}$, so the final charge on inner shell, $q' = -(3q)\left(\frac{a}{b}\right)$.

Q. **There are three thin shells A, B and C shown in figure with their radii and charge configurations.**
Find the charge configurations on the shells in the following cases when
(a) A is earthed.
(b) C is earthed.
(c) When shell A and C are earthed.
(d) When A and C are connected by a conducting wire and B is earthed.



Sol. (a) When A is earthed, new charge with appear on shell A. Let charge on shell A is q_A.

Now, write electric potential of A due to all the shells.

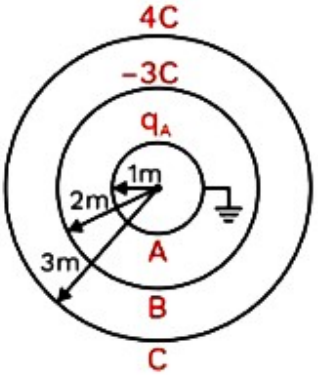
$$V_A = \frac{Kq_A}{1} + \frac{K(-3)}{2} + \frac{K(4)}{3}$$

$$V_A = \frac{Kq_A}{1} - \frac{3K}{2} + \frac{4K}{3}$$

Since A is earthed, so $V_A = 0$.

$$\frac{Kq_A}{1} - \frac{3K}{2} + \frac{4K}{3} = 0$$

$$\boxed{q_A = +\frac{1}{6} \text{ coulomb}}$$

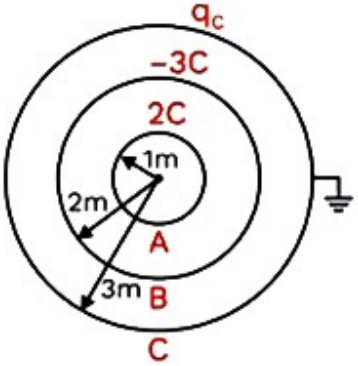


(b) When C is earthed, let the new charge appearing on shell 'C' is q_C.

$$V_C = \frac{K(2C)}{3} + \frac{K(-3C)}{3} + \frac{K(q_C)}{3}$$

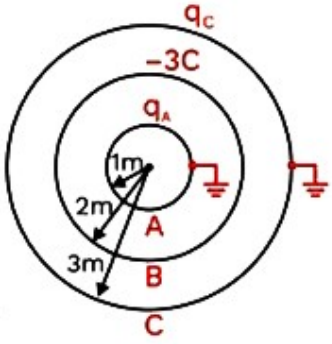
Since 'C' is earthed, so $V_C = 0$.

After solving we get,



$\boxed{q_C = 1 \text{ coulomb}}$.

(c) When shell A and C are earthed.



Let new charges appearing on shell A and C are q_A and q_C respectively

According to given condition:

$$V_A = \frac{Kq_A}{1} + \frac{K(-3C)}{2} + \frac{Kq_C}{3}$$

Since shell 'A' is earthed so potential of A is equal to zero.

$$q_A - \frac{3}{2} + \frac{q_C}{3} = 0$$

$$3q_A + q_C = \frac{9}{2} \qquad \ldots(1)$$

Now write potential at 'C'.

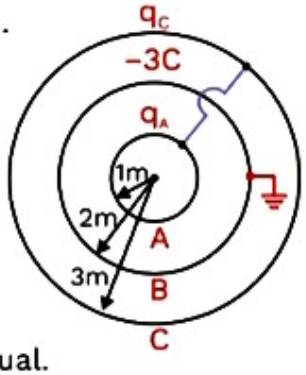
$$\frac{Kq_A}{3} + \frac{K(-3)}{3} + \frac{Kq_C}{3} = 0$$

$$q_A + q_C = 3 \qquad \ldots(2)$$

After solving equation −1 and equation−2

We get ; $\boxed{q_A = \left(\frac{3}{4}\right)C \quad ; \quad q_C = \left(\frac{9}{4}\right)C}$

(d) When A and C are connected by a wire and B is earthed.



According to charge conservation principle:

$q_A + q_C = 6 \qquad \ldots(1)$

Potential at 'B':

$$\frac{q_A}{2} + \frac{q_B}{2} + \frac{6 - q_A}{3} = 0$$

$$q_A + 3q_B = -12 \qquad \ldots(2)$$

Since A and C are connected so potential of A and C are equal.

$$V_A = V_C$$

$$\frac{Kq_A}{1} + \frac{Kq_B}{2} + \frac{Kq_c}{3} = \frac{Kq_A}{3} + \frac{Kq_B}{3} + \frac{K(6 - q_A)}{3}$$

$$4q_A + 3q_B + 12 = 2q_B + 12$$

$$q_B = 4q_A$$

Now use second equation:

$$q_A + 3(-4q_A) = -12$$

$$V_A = V_C$$

$$\frac{Kq_A}{1} + \frac{Kq_B}{2} + \frac{Kq_c}{3} = \frac{Kq_A}{3} + \frac{Kq_B}{3} + \frac{K(6 - q_A)}{3}$$

$$4q_A + 3q_B + 12 = 2q_B + 12$$

$$q_B = -4q_A \quad \text{.....(3)}$$

Now use second equation:

$$q_A + 3(-4q_A) = -12$$

$$\boxed{q_A = \frac{12}{11}\text{coulomb}}.$$

From equation (3) ; $q_B = -4\left(\frac{12}{11}\right) = \frac{-48}{11}C$

From equation (1) ; $q_c = 6 - q_A$

$$= 6 - \frac{12}{11} = \frac{66 - 12}{11}$$

$$= \frac{54}{11}C$$

Electric field near the surface of conductor

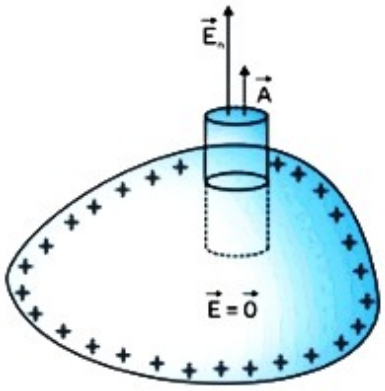
To find the electric field just outside a conductor, consider the Gaussian pillbox drawn in figure. Using Gauss's law,

$$\phi_E = \oint \vec{E} \cdot \overrightarrow{dA}$$

$$\frac{q_{enc}}{\varepsilon_0} = E_n A + (0) \cdot A$$

$$\Rightarrow \quad \frac{\sigma A}{\varepsilon_0} = E_n A$$

$$\Rightarrow \quad E_n = \frac{\sigma}{\varepsilon_0}$$



Electrostatic pressure at the surface of a charged conductor

All the charge given to a conductor resides on its outer surface. These charges on the outer surface repel each other due to their similar nature.

So, each charge experiences an outward force due to the repulsion from other charges.

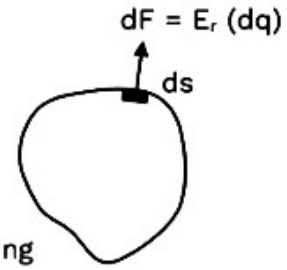
This repulsive force causes a pressure on the surface of the conductor known as electrostatic pressure.



Consider an elemental area 'ds' on the surface of the conductor.

Force acting on the elemental area due to remaining charges,

$$dF = \begin{pmatrix} \text{Electric field at} \\ \text{that place due to} \\ \text{remaining charges} \end{pmatrix} \begin{pmatrix} \text{Charge of} \\ \text{the small} \\ \text{element} \end{pmatrix}$$



Consider E_r = electric field near the elemental area due to remaining charges.

And, charge on elemental area = $dq = \sigma\, ds$.

Then, $dF = (E_r)\,(dq) = (E_r)\,(\sigma ds)$

So, pressure acting on elemental area,

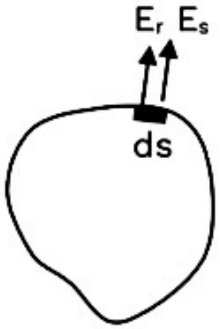
$$P = \frac{dF}{ds} = \frac{(E_r)(\sigma ds)}{ds} \Rightarrow P = (E_r)\,(\sigma). \quad \ldots(1)$$

We need to find the value of E_r (electric field at that position due to the remaining charges).

Let E_s = electric field due to charge on elemental area.

For a point lying just outside the surface, E_r and E_s acts normally outwards.

So, total electric field just outside the surface = $E_s + E_r$



It is also known that, electric field just outside the conductor surface = $\frac{\sigma}{\varepsilon_o}$.

$$\Rightarrow E_s + E_r = \frac{\sigma}{\varepsilon_o} \quad \ldots(2)$$

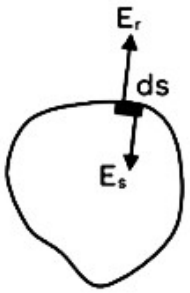
For a point just inside the surface of conductor, E_s is normally inwards while E_r is normally outwards.

So, total electric field just inside the metal surface = $E_r - E_s$.

Also, total electric field inside a conductor = 0

So, $E_r - E_s = 0 \Rightarrow E_r = E_s \quad \ldots(3)$

From eqn. (2) and eqn. (3), we can say that



$$2E_r = \frac{\sigma}{\varepsilon_0} \Rightarrow E_r = \frac{\sigma}{2\varepsilon_0}$$

Now, we can easily find the pressure from equation (1),

$$P = (E_r)(\sigma) = \frac{\sigma}{2\varepsilon_0}(\sigma) = \frac{\sigma^2}{2\varepsilon_0}$$

So, electrostatic pressure at the surface of the conductor, $P = \frac{\sigma^2}{2\varepsilon_0}$.

where, σ = local surface charge density.